Ancestral Skies Collaboration

MATH3202 Operations Research and Mathematical Planning
Assignment 2

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Section A: Mathematical Formulation

Sets

$Skills$	Skills
$Tasks$	Tasks
$Staff$	Staff members
$Days$	Days in a week
$forbiddenPairs$	Pairs of Staff not to be assigned to the same Task

Data

$title_j$	Title of task $j \in Tasks$
$skill_j$	skill $s \in Skills$ required for task $j \in Tasks$
$staff_j$	number of staff required for task $j \in Tasks$
$duration_j$	duration of task $j \in Tasks$
day_j	the day $d \in Days$ that task $j \in Tasks$ must be completed
SS_{ij}	Skill scores for staff $i \in Staff$ at skill required for Task $j \in Tasks$
fP_{i_1,i_2}	Forbidden staff pairing: staff i_1 and i_2 cannot be assigned to the same task, where $i_1, i_2 \in Staff$
$maxWeeklyHours$	maximum number of hours each staff member can work in a week
$maxDailyHours$	maximum number of hours each staff member can work in a day
$minRestDays$	minimum number of days each staff member must have off work in a week
$maxDaysWorked$	maximum number of days each staff member can work in a week (7 days in a week - $minRestDays$)
$nightTask$	task s.t. if this task is allocated to a staff member, they can perform no other task in the same day
$trainingSlots$	number of staff members that can receive extra training at the start of a work week
$trainingSkillIncrease$	number of additional skill points that staff gain after training before the week starts

Variables

$$X_{ij} = \begin{cases} 1 & \text{if staff } i \in Staff \text{ is assigned to task } j \in Tasks \\ 0 & \text{otherwise} \end{cases}$$

$$Y_{id} = \begin{cases} 1 & \text{if staff } i \in Staff \text{ is working on day } d \in Days \\ 0 & \text{otherwise} \end{cases}$$

$$Z_{is} = \begin{cases} 1 & \text{if staff } i \in Staff \text{ is trained in skill } s \in Skills \\ 0 & \text{otherwise} \end{cases}$$

$$W_{ij} = \begin{cases} 1 & \text{if staff } i \in Staff \text{ is assigned and is trained in skill required for task } j \in Tasks \\ 0 & \text{otherwise} \end{cases}$$

Objective

$$\max \sum_{j \in Tasks} \sum_{i \in Staff} (X_{ij} \cdot SS_{ij} + W_{ij} \cdot trainingSkillIncrease)$$

Constraints

Max Hours Weekly Constraint

$$\sum_{j \in Tasks} X_{ij} \cdot duration_j \leq maxWeeklyHours \quad \forall i \in Staff$$

Number of Required Staff per Task Constraint

$$\sum_{i \in Staff} X_{ij} = staff_j \quad \forall j \in Tasks$$

Forbidden Pairs Constraint (Communication 7)

$$\sum_{i \in fP_{i_1,i_2}} X_{ij} \leq 1 \quad \forall fP_{i_1,i_2} \in forbiddenPairs, \quad \forall j \in Tasks$$

Max Hours Daily Constraint (Communication 8)

$$\sum_{\substack{j \in Tasks \\ day_j = d}} X_{ij} \cdot duration_j \leq maxDailyHours \quad \forall d \in Days,\ i \in Staff$$

Days Off Constraint (Communication 8)

$$\sum_{d \in Days} Y_{id} \leq maxDaysWorked \quad \forall i \in Staff$$

Linking Days Worked to Tasks Performed Constraint (Communication 8)

$$M \cdot Y_{id} \geq \sum_{\substack{j \in Tasks \\ day_j = d}} X_{ij} \quad \forall d \in Days, \quad \forall i \in Staff$$

where $M = 1000000$

Nightly Telescope Operation Constraint (Communication 9)

$$\sum_{\substack{j \in Tasks \\ title_j \neq nightTask,\ day_j = d}} X_{ij} \leq M \cdot (1 - \sum_{\substack{j \in Tasks \\ title_j = nightTask,\ day_j = d}} X_{ij}) \quad \forall d \in Days, \quad \forall i \in Staff$$

where $M = 1000000$

Training Slots Constraint (Communication 10)

$$\sum_{s \in Skills} \sum_{i \in Staff} Z_{is} \leq trainingSlots$$

One Training per Staff Constraint (Communication 10)

$$\sum_{s \in Skills} Z_{is} \leq 1 \quad \forall i \in Staff$$

Linking variable W to variables Z, X (Communication 10)

$$W_{ij} \leq X_{ij} \quad \forall j \in Tasks, \quad \forall i \in Staff$$

$$W_{ij} \leq Z_{i,skill_j} \quad \forall j \in Tasks, \quad \forall i \in Staff$$

$$W_{ij} \geq X_{ij} + Z_{i,skill_j} - 1 \quad \forall j \in Tasks, \quad \forall i \in Staff$$

Non-negativity Constraints

$$X_{ij} \geq 0 \quad \forall i \in Staff, \quad \forall j \in Tasks$$

$$Y_{id} \geq 0 \quad \forall i \in Staff, \quad \forall d \in Days$$

$$Z_{is} \geq 0 \quad \forall i \in Staff, \quad \forall s \in Skills$$

Section A: Code

Please see the attached Python (.py) file containing the code solution for the Ancestral Skies Collaboration.

Section B: Report to Client

Please see the part B file for the report to the client.